WORKED SOLUTION

STEP III 2005, Question 8

Pure • Complex Numbers and Circles

STEP III 2005, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q8

The key idea

Every length in the Argand diagram can be turned into algebra with $|z|^2 = zz^*$. Once the first identity is in hand, a right angle becomes Pythagoras and a tangent becomes a right angle between the tangent and the radius. So "OA touches the circle" turns into a single equation, which fixes the radius squared as $cc^* - aa^*$. After that, both "P on the circle" and "P' on the circle" reduce to the same equation.

Solution

The identity and the right angle

Since $|z|^2 = zz^*$ and conjugation respects subtraction,

$$\begin{aligned} |a-c|^2 &= (a-c)(a-c)^* = (a-c)(a^*-c^*) \\ &= aa^* + cc^* - ac^* - ca^*, \end{aligned}$$

as required.

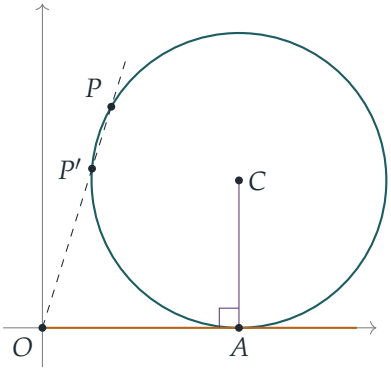
The sides of triangle OAC have lengths $OA = |a|$, $OC = |c|$ and $AC = |a-c|$, all non-zero because a, c are non-zero and distinct. By Pythagoras' theorem and its converse, the angle at A is a right angle if and only if $OC^2 = OA^2 + AC^2$, that is,

$$cc^* = aa^* + (aa^* + cc^* - ac^* - ca^*) \iff 2aa^* = ac^* + ca^*.$$

Every step here is reversible, so the "if and only if" holds.

(The converse of Pythagoras also rules out a degenerate "triangle". If O, A, C were collinear then one of the three lengths would equal the sum of the other two, and that can't happen alongside $OC^2 = OA^2 + AC^2$ when all three lengths are positive.)

The tangent circle



A tangent is perpendicular to the radius at the point of contact. So OA being tangent at A means that angle OAC is a right angle, and by the first part

$$2aa^* = ac^* + ca^*. \qquad (*)$$

The radius r satisfies $r^2 = |a-c|^2 = aa^* + cc^* - (ac^* + ca^*) = cc^* - aa^*$, using $(*)$. (That's just Pythagoras again: $r^2 = OC^2 - OA^2$.)

Write $k = bb^* > 0$ and $s = abc^* + a^*b^*c$. Then

$$|ab-c|^2 = (ab-c)(a^*b^* - c^*) = aa^*k - s + cc^*,$$
$$\left|\frac{a}{b^*} - c\right|^2 = \left(\frac{a}{b^*} - c\right)\left(\frac{a^*}{b} - c^*\right) = \frac{aa^*}{k} - \frac{ac^*}{b^*} - \frac{a^*c}{b} + cc^*.$$

Multiplying the second by $k = bb^*$ gives

$$k\left|\frac{a}{b^*} - c\right|^2 = aa^* - s + k\,cc^*. \qquad (**)$$

P on the circle. $|ab-c|^2 = cc^* - aa^*$ holds if and only if

$$aa^*k - s + cc^* = cc^* - aa^* \iff aa^*(k+1) = s.$$

P' on the circle. $|a/b^* - c|^2 = cc^* - aa^*$ holds if and only if (multiplying by $k \neq 0$ and using $(**)$)

$$aa^* - s + k\,cc^* = k\,cc^* - k\,aa^* \iff aa^*(k+1) = s.$$

The two conditions are identical, so P lies on the circle if and only if P' does.

The converse

Now drop the tangency assumption. Suppose the circle has centre C and passes through A, so $r^2 = |a-c|^2 = aa^* + cc^* - ac^* - ca^*$. Suppose also that both ab and a/b^* lie on it, where $k = bb^* \neq 1$. Using the same expansions,

$$aa^*k - s + cc^* = r^2,$$
$$aa^* - s + k\,cc^* = kr^2 \qquad \text{(from } (**)\text{)}.$$

Subtracting the second equation from the first eliminates s:

$$(k-1)\,aa^* - (k-1)\,cc^* = (1-k)\,r^2.$$

Ask yourself Why does the question insist on $bb^* \neq 1$?

If $bb^* = 1$ then $b^* = 1/b$, so $a/b^* = ab$ and the two points coincide. "Both points on the circle" is then a single condition, which certainly doesn't force tangency. In our algebra this is exactly the step where we divide by $k - 1 \neq 0$:

$$r^2 = cc^* - aa^*.$$

Comparing with $r^2 = aa^* + cc^* - ac^* - ca^*$ gives $2aa^* = ac^* + ca^*$. By the first part, angle OAC is a right angle. So OA is perpendicular to the radius CA at the point A of the circle, which means OA is a tangent to the circle.

Checks

- There's a neat geometric reason behind the result. Since $\dfrac{a}{b^*} = \dfrac{ab}{bb^*}$, the point P' is P scaled by the positive real $1/|b|^2$. So O, P, P' are collinear, with $OP \cdot OP' = |a|^2 = OA^2$, which is exactly the tangent–secant theorem (see the diagram).
- Numerically, take $a = 2$, $c = 2 + 1.5\mathrm{i}$ (so $r^2 = 2.25 = |c|^2 - |a|^2$), and $P = 0.701 + 2.25\mathrm{i}$ on the circle. Then $b = P/2$ and $P' = a/b^* \approx 0.505 + 1.620\mathrm{i}$, with $|P' - c|^2 \approx 2.250$.
- A computer checked the identity and the tangent case on random a, b, c. It also took 1000 random non-tangent circles with $|b| \neq 1$: in every case, P on the circle forced P' off it.

Common slips

- Writing $(a-c)^* = a^* - c^*$ correctly but then expanding $|a-c|^2$ as $(a-c)^2$. The modulus squared needs the conjugate.
- In the converse, assuming tangency (and so $r^2 = cc^* - aa^*$) at the start. That is what you have to prove. Start from $r = |a - c|$ only.
- Dividing by $1 - bb^*$ without saying why it's non-zero.

Beyond STEP

Since $|a|^2/(ab)^* = aa^*/(a^*b^*) = a/b^*$, the point P' is the image of P under $z \mapsto |a|^2/z^*$. This map is *inversion* in the circle $|z| = |a|$: it sends each point to the point on the same ray from O at distance $|a|^2/OP$. This question shows that circles orthogonal to the circle of inversion are mapped to themselves. Inversion and the Möbius transformations it

generates are central to first-year complex analysis and geometry. Needham's *Visual Complex Analysis* has a beautiful chapter-length treatment.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.